Project 3: Applied Demonstration of Regularization

MA289: Intro to Independent Study in Math

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Problem

The challenge of using linear models in many cases is that, given a number of features, some of them matter and others do not. As a result, linear models often overfit and overcomplicate the model to compensate for noise, even when that noise has no real significance. For the real dataset, we have ten features, many of which are mathematically dependent on each other. Because of this multicollinearity, and other factors that cause the model to overfit, the model fails to do well on new data. To solve this problem, we have to use regularization to minimize the noise for the model. The datasets selected are intentionally designed to stress test Ordinary Least Squares, which has no regularization.

Explanation of the Modeling Methods

Ordinary Least Squares (OLS) is a baseline linear model that minimizes squared error and treats every feature equally. Ridge regularization shrinks the coefficients of features that it determines to be less correlated, but is under the limitation that it cannot eliminate any feature. Lasso regularization is similar to Ridge in that it minimizes the impact associated with features that it determines to be noise, but the difference is that it completely zeroes them out. Lastly, the comparison of Best Subset Selection is used, by using AIC and BIC. The difference is that BIC penalizes complexity of the model more harshly than AIC, so it tends to select sparser models.

To choose lambda, the method of five-fold cross validation is used. It picks from a selection of 200 candidate values on a log scale.

Description of Pathological Synthetic Dataset

The synthetic dataset is 150 samples and 50 features, only 5 of which are not pure noise and have known coefficients. Because of the low number of informative features, it stresses OLS, because it considers all 50 features equally.

Description and Justification of Real Dataset

The real dataset used is a sklearn diabetes dataset, comprised of 442 patients with 10 clinical features. Those features are: age, sex, BMI, BP, and six other serum measurements that are mostly mathematically correlated to each other. It challenges OLS because the features are highly correlated to each other, and others have virtually no impact on diabetes.

Mathematical Formulation of the Method

OLS objective:

$$\min_{\beta} \; \|y - X\beta\|_2^2$$

Ridge objective:

$$\min_{\beta} \; \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$$

where $\|\beta\|_2^2$ is the sum of squared coefficients.

Lasso objective:

$$\min_{\beta} \; \|y - X\beta\|_2^2 + \lambda\|\beta\|_1$$

where $\|\beta\|_1$ is the sum of absolute coefficients.

Best Subset Selection criteria:

$$\text{AIC} = -2\log(L) + 2k \qquad \text{BIC} = -2\log(L) + k\log(n)$$

where L is the model likelihood, k is the number of features, and n is the sample size.

Experimental Results and Comparison

For both the synthetic dataset and the real one, Lasso regularization was found to be the superior model, by a measure of the Mean Squared Error (MSE), but not by much. For the real dataset, Lasso had an MSE of 2801, as opposed to Best Subset's 2846 (AIC and BIC were the same), Ridge's 2863, and OLS's 2900. Best Subset just underperformed Lasso, but was able to do it with 6 features compared to Lasso's 7, and OLS and Ridge's 10, indicating a much higher efficiency.

Discussion of Limitations and Practical Implications

Although Lasso was found to be the best model to be generalized to new data, it is important to consider the increased efficiency of Best Subset Selection. Although Lasso zeroed out 30% of all features, Best Subset zeroed out 40%. With a large dataset, this difference would make you consider using Best Subset to model the data, so you could process the model faster.